

AWS



FUNDAMENTALS

BUILD THE FOUNDATION. MASTER THE CLOUD.

“AWS looks complicated – until you understand the fundamentals.”



COVERED IN THESE NOTES

01

CLOUD & AWS BASICS



- What is Cloud Computing?
- Types of Cloud
- AWS Overview
- Benefits of AWS

02

GLOBAL INFRASTRUCTURE



- Regions
- Availability Zones
- Edge Locations
- Local Zones
- How AWS Global Infra Works

03

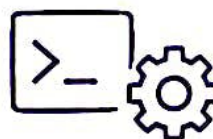
IAM & ACCESS



- Users, Groups, Roles
- Policies
- Permissions
- Best Practices

04

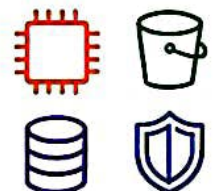
AWS TOOLS & IaC



- AWS Management Console
- AWS CLI
- SDKs & APIs
- Infrastructure as Code (IaC)
- Automation

05

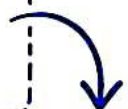
CORE AWS SERVICES



- Compute
- Storage
- Database
- Networking
- Security
- Monitoring
- And More...



Want the interview-ready PDF?
DM me “AWS” and I’ll share it.



ONE CONCEPT AT A TIME. ONE STEP CLOSER.



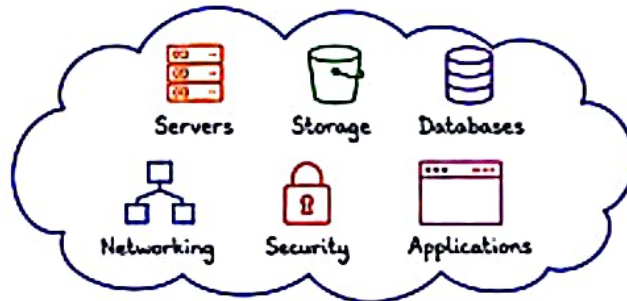
@NeerajSingh-DevOps

WHAT IS CLOUD COMPUTING & AWS?

AWS (Amazon Web Services) is a cloud platform that provides **on-demand** computing resources over the internet with high availability, scalability and **pay-as-you-go** pricing.

1 WHAT IS CLOUD COMPUTING?

- Cloud Computing means using computing resources over the internet.
- These resources can be servers, storage, databases, networking, security, applications and more.
- You use only what you need and pay only for what you use.



2 TRADITIONAL INFRASTRUCTURE vs CLOUD

TRADITIONAL (ON-PREMISES)

- Buy Physical Servers
- Setup Data Center
- Manage Hardware
- Install OS & Software
- Configure Network
- Handle Security
- Maintain & Upgrade
- High Cost Upfront



You Manage Everything

VS

CLOUD (AWS)

- No need to buy Hardware
- AWS manages infrastructure
- Provision in minutes
- Scale up or down easily
- Built-in Security
- High Availability
- Pay for what you use
- Low Cost Upfront

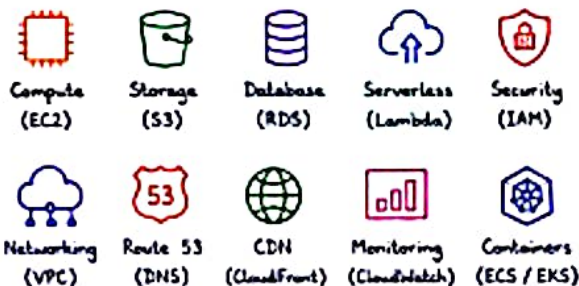


AWS Manages Infrastructure

3 WHAT IS AWS?

- AWS is the world's most comprehensive and widely adopted cloud platform.
- It offers 200+ fully featured services from data centers globally.
- Used by startups, enterprises, governments and millions of customers worldwide.

AWS SERVICE CATEGORIES (Examples)



4 WHY DO COMPANIES USE AWS?

- ✓ Scalability → Increase or decrease resources easily.
- ✓ High Availability → Built for reliability and uptime.
- ✓ Global Infrastructure → Deploy worldwide.
- ✓ Pay-as-you-go → Pay only for what you use.
- ✓ Security → Industry-leading security services.
- ✓ Managed Services → AWS manages infrastructure.
- ✓ Innovation → Continuous new services and features.
- ✓ Automation → Integrate with DevOps tools easily.

5 SERVICE MODELS

IaaS (Infrastructure as a Service)	Rent infrastructure like servers, storage, networking. Example: EC2, S3, VPC
PaaS (Platform as a Service)	Use platform to build, run and manage applications without managing infrastructure. Example: Elastic Beanstalk, RDS
SaaS (Software as a Service)	Use ready-to-use software over the internet. Example: Gmail, Salesforce, Office 365

6 PAY-AS-YOU-GO MODEL



★ Great for:

- Startups
- Projects with unpredictable workloads
- Cost optimization

7 INTERVIEW ANGLE (BASICS)

- ★ What is AWS?
- ★ What is Cloud Computing?
- ★ Differences between Traditional Infrastructure and Cloud?
- ★ What are the benefits of AWS?
- ★ What are IaaS, PaaS and SaaS?
- ★ What is Pay-as-you-go model?



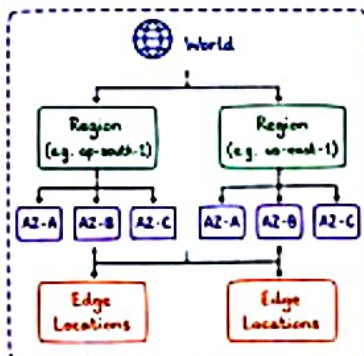
★ **TIP:** Think of AWS as a huge toolbox in the cloud. You pick the right tools (services) and build your solution!

ACCOUNT, BILLING & PRICING

Before using AWS, understand account setup, billing and pricing to manage costs and avoid surprises.

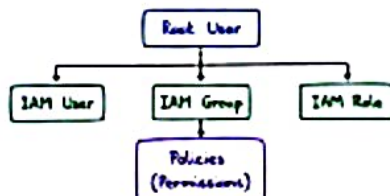
1 AWS GLOBAL INFRASTRUCTURE

- AWS has data centers all over the world.
- Made up of Regions, Availability Zones (AZs) and Edge Locations.
- You choose the Region closest to your users.



2 AWS ACCOUNT STRUCTURE

- Root user has full access.
- Create IAM users for daily tasks.
- Use groups and roles to manage permissions.
- Enable MFA for security.

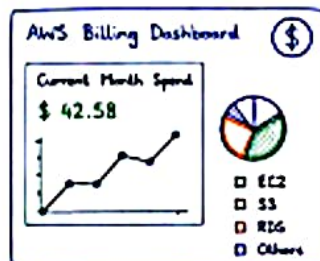


BEST PRACTICES

- ✓ Don't use Root user for daily tasks.
- ✓ Enable MFA.
- ✓ Follow Least Privilege principle.
- ✓ Use IAM Roles for AWS services.

3 BILLING DASHBOARD

- View charges, services used and trends.
- Monitor costs in real-time.
- Set Budget and Alerts.
- Download invoices.



4 PRICING MODELS

- Pay-As-You-Go - Pay only for what you use.
- Free Tier - Free usage for 12 months.
- Reserved Instances - Commit for 1 or 3 years, save more.
- Savings Plans - Flexible pricing for compute usage.

MODEL	DESCRIPTION
Pay-As-You-Go	No upfront cost. Pay for usage.
Free Tier	Free for 12 months for eligible services.
Reserved Instances	Commit 1 or 3 years. Discounted pricing.
Savings Plans	Flexible commitment. Lower prices.

5 COST OPTIMIZATION TIPS



Monitor with AWS Cost Explorer



Set Budget and Alerts



Right-size Resources



Stop Unused Resources



Use Reserved Instances or Savings Plans



Review & Optimize Regularly

6 COMMON ACCOUNT TASKS

TASK	HOW TO DO	SERVICE / TOOL	WHY IT'S IMPORTANT
Create IAM User	Go to IAM → Users → Add user	IAM	For secure access without using Root user.
Attach Policies	Add user to group or attach policy directly	IAM	To give required permissions only.
Enable MFA	Security Credentials → Activate MFA	IAM	Adds extra layer of security.
Set Budget	Go to Billing → Budgets → Create Budget	AWS Budgets	Helps track and control spending.
Set Alerts	Create budget → Add alerts → Choose SNS / Email	AWS Budgets Amazon SNS	Get notified when cost reaches limit.
Review Costs	Open Cost Explorer → Analyze usage and cost	AWS Cost Explorer	Understand usage pattern and optimize.
Download Invoice	Billing Dashboard → Download Invoice	Billing Console	For records and compliance.

7 KEY TAKEAWAYS

- ✓ AWS is a global cloud platform with Regions and AZs.
- ✓ Use IAM users, groups, roles and MFA for security.
- ✓ Monitor your costs and set budgets to avoid over-spending.
- ✓ Choose the right pricing model for your needs.
- ✓ Always follow best practices for cost and security.

Good planning = Security + Cost Savings + Better Performance!

8 INTERVIEW ANGLE

- ★ What is the difference between Root user and IAM user?
- ★ What is MFA and why is it important?
- ★ How does AWS pricing work?
- ★ What are AWS Free Tier limits?
- ★ How will you monitor and optimize AWS costs?

Think from Security, Cost and Efficiency perspective!

9 NEXT PAGE PREVIEW

Next, we will learn about core AWS services like Compute, Storage, Database, Networking and more.



SECURITY, IAM & MANAGEMENT TOOLS



Security is the foundation of AWS. Follow the principle of Least Privilege - Grant only what is needed.

1 AWS SECURITY BEST PRACTICES

- Use IAM to control access.
- Enable MFA for all users.
- Follow Least Privilege access.
- Use Security Groups, NACLs and KMS to secure resources.
- Encrypt data at rest and in transit.
- Monitor and log everything.
- Regularly review permissions and access.

SHARED RESPONSIBILITY MODEL

AWS (Security OF the Cloud)
Infrastructure, Hardware, Networking, Facilities, Global Services



YOU (Security IN the Cloud)
Data, IAM, OS, Applications, Network Config, Access Management



2 IAM (IDENTITY & ACCESS MANAGEMENT)

IAM helps you securely control access to AWS services and resources.

Key Components:

- Users** - Individuals who access AWS.
- Groups** - Collection of users.
- Roles** - Permissions without users (for EC2, Lambda, etc.)
- Policies** - JSON documents that define permissions.
- Access Keys** - For programmatic access (Use with care)

IAM POLICY EXAMPLE

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:ListBucket"
      ],
      "Resource": [
        "arn:aws:s3:::my-bucket"
      ]
    }
  ]
}
```

3 AWS MANAGEMENT TOOLS

SERVICE	DESCRIPTION	USE CASE
CloudWatch	Monitor resources, logs, metrics and alarms	Monitoring, Alarms, Dashboards
CloudTrail	Tracks API calls and user activity	Audit, Compliance, Troubleshooting
AWS Config	Records and evaluates resource configuration	Compliance, Change Tracking
Trusted Advisor	Provides best practice recommendations	Cost Optimization, Security, Fault Tolerance
Systems Manager	Manage EC2, automate tasks, patching, run commands	Automation, Patch Management
AWS Health Dashboard	Status AWS service health and events	Stay informed about AWS service status

4 ENCRYPTION IN AWS

- Protect your data using encryption.
- Types:

Encryption at Rest



Data stored in S3, EBS, RDS, DynamoDB, etc.

Example: S3 Bucket SSE-S3 / SSE-KMS

Encryption in Transit



Data moving between services or over internet.

Example: HTTPS, SSL/TLS

KMS (Key Management Service)

- Manage encryption keys
- Used for encrypting data and controlling access to keys.



Tip: Always use AWS managed services (IAM, KMS, CloudTrail) to build a secure and compliant environment.

5 AWS CLI & SDK

AWS CLI (Command Line Interface)

```
>>>
$ aws s3 ls
2024-05-01 my-bucket/
2024-05-01 another-bucket/
```

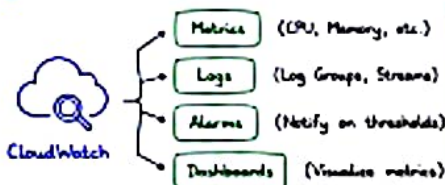
- Manage AWS services from your terminal.
- Automate tasks and scripts.
- Great for DevOps and Automation.

AWS SDK (Software Development Kit)

```
000
import boto3
s3 = boto3.client('s3')
response = s3.list_buckets()
for bucket in response['Buckets']:
    print(bucket['Name'])
```

- Use AWS services in your applications.
- Available for many languages (Python, Java, JavaScript, Go, etc.).

6 CLOUDWATCH OVERVIEW



EXAMPLE USE CASE

- Set an alarm if CPU usage > 80%
- Get notified via Email / SNS.



7 KEY TAKEAWAYS

- ✓ Security comes first - Always!
- ✓ Use IAM and MFA.
- ✓ Understand Shared Responsibility Model
- ✓ Monitor, Log and Audit everything
- ✓ Encrypt your data.
- ✓ Use AWS Management Tools to simplify operations.



INTERVIEW QUESTIONS (BASICS)

- 1 What is the difference between IAM user and IAM role?
- 2 What is MFA and why is it important?
- 3 What is the Shared Responsibility Model in AWS?
- 4 How does AWS CloudTrail help in security?
- 5 What is the difference between Security Group and NACL?
- 6 What is KMS and how does it work?
- 7 What are AWS Management Tools and their uses?
- 8 How can you monitor AWS resources?
- 9 What is the purpose of AWS Config?
- 10 What are some AWS Security Best Practices?

Understand Security deeply. It is key for any Cloud Engineer!



AWS FUNDAMENTALS

PAGE 4 OF 5

CORE AWS SERVICES OVERVIEW



AWS provides 200+ services. Below are some core services you should know as a beginner.

1 COMPUTE

SERVICE	WHAT IT DOES
EC2	Virtual servers in the cloud. Fully customizable.
Elastic Beanstalk	Easily deploy and manage applications.
Lambda	Run code without managing servers. Pay only for usage.
Auto Scaling	Automatically adjust EC2 capacity based on demand.

2 STORAGE

SERVICE	WHAT IT DOES
S3	Object storage for files, images, videos, backups.
EBS	Block storage for EC2 instances.
EFS	Managed file storage for multiple instances.
Storage Gateway	Hybrid storage between on-premises and AWS.

3 DATABASE

SERVICE	WHAT IT DOES
RDS	Managed relational database (MySQL, PostgreSQL, etc.).
DynamoDB	NoSQL database. Fast and highly scalable.
Redshift	Data warehouse for analytics.
ElastiCache	In-memory caching service (Redis, Memcached).

4 NETWORKING & CONTENT DELIVERY

SERVICE	WHAT IT DOES
VPC	Isolated virtual network in AWS.
Route 53	Scalable DNS web service.
CloudFront	Content Delivery Network (CDN) for fast delivery.
ELB	Distribute traffic across instances.

5 SECURITY

SERVICE	WHAT IT DOES
IAM	Manage users, groups, roles and permissions.
KMS	Manage encryption keys securely.
Security Groups	Act as virtual firewall for your resources.
Secrets Manager	Store and manage secrets like passwords, API keys.

6 MANAGEMENT & MONITORING

SERVICE	WHAT IT DOES
CloudWatch	Monitor resources, logs, metrics and set alarms.
CloudTrail	Track API calls and user activity.
AWS Config	Track resource configuration changes.
Trusted Advisor	Best practice checks and recommendations.

7 ANALYTICS

SERVICE	WHAT IT DOES
Athena	Query data in S3 using standard SQL.
EMR	Managed Hadoop framework for big data.
QuickSight	Business intelligence and data visualization.
Kinesis	Real-time data streaming and processing.

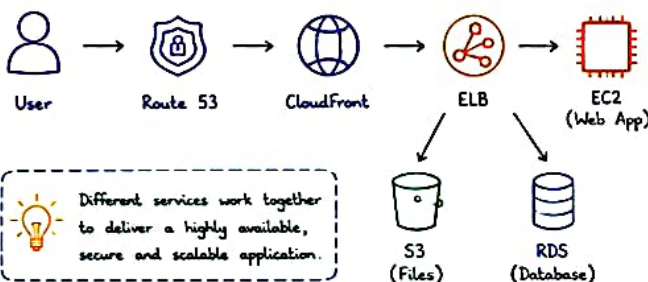
8 INTEGRATION

SERVICE	WHAT IT DOES
Step Functions	Orchestrate workflows of AWS services.
EventBridge	Build event-driven applications.
SQS	Message queue for decoupling applications.
SNS	Publish messages and send notifications.

9 DEVELOPER TOOLS

SERVICE	WHAT IT DOES
CodeCommit	Managed Git repositories.
CodeBuild	Build and test your code automatically.
CodeDeploy	Automate code deployments.
CodePipeline	Automate your CI/CD pipeline.

HOW SERVICES WORK TOGETHER (EXAMPLE)



QUICK RECAP

- ★ Compute → EC2, Lambda, Elastic Beanstalk
- ★ Storage → S3, EBS, EFS
- ★ Database → RDS, DynamoDB, Redshift
- ★ Networking → VPC, Route 53, CloudFront, ELB
- ★ Security → IAM, KMS, Security Groups
- ★ Monitoring → CloudWatch, CloudTrail
- ★ Developer Tools → CodeCommit, CodeBuild, CodePipeline
- ★ Integration → SQS, SNS, EventBridge

KEY TAKEAWAY



You don't need to learn all services at once. Start with the core ones and build projects. Hands-on practice is the best way to understand AWS services!



NEXT PAGE PREVIEW

- AWS Free Tier, Support Plans & Pricing
- Final Key Takeaways & Resources



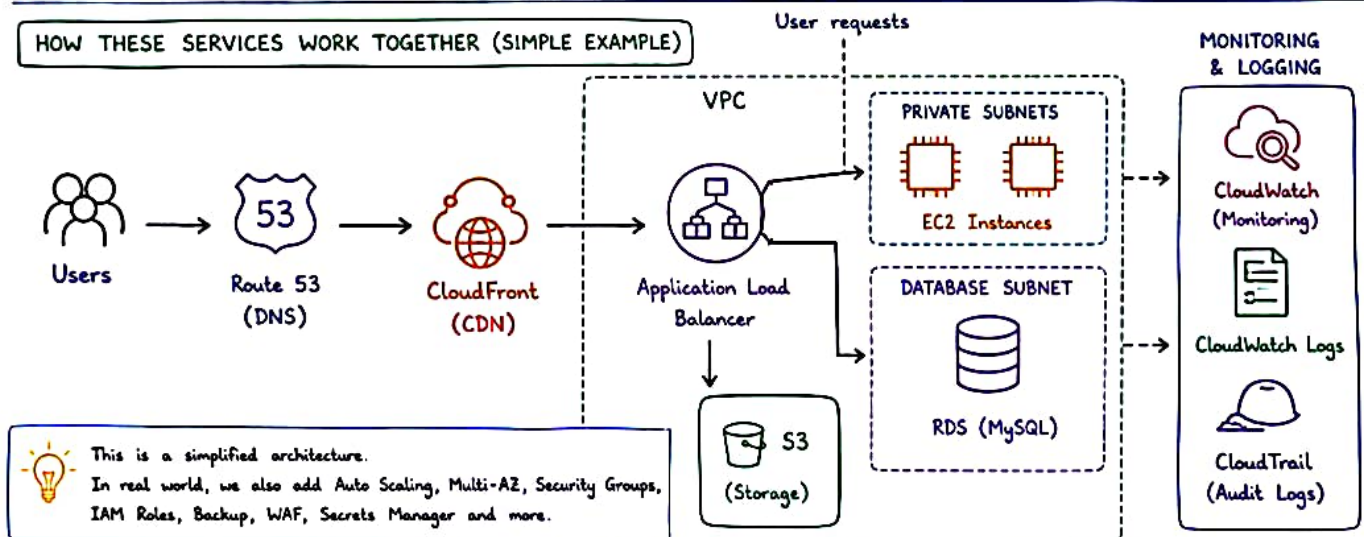
AWS CORE SERVICES MAP

AWS provides **200+ services**. Here are the most commonly used core services.

COMPUTE  Run virtual servers in cloud. Example: EC2	STORAGE  Store and retrieve any amount of data. Example: S3	DATABASE  Managed relational or NoSQL databases. Example: RDS, DynamoDB	NETWORKING  Isolated network for your resources. Example: VPC	DNS  Domain name system. Example: Route 53
CDN  Deliver content faster globally. Example: CloudFront	SERVERLESS  Run code without managing servers. Example: Lambda	CONTAINERS  Run and manage containerized apps. Example: ECS / EKS	MONITORING  Monitor resources and applications. Example: CloudWatch	SECURITY  Manage access, encrypt and protect. Example: IAM, KMS

★ These services work together to build highly available, secure and scalable applications.

HOW THESE SERVICES WORK TOGETHER (SIMPLE EXAMPLE)



KEY TAKEAWAYS

- ✓ AWS gives infrastructure on demand over the internet.
- ✓ Global Infrastructure helps us build highly available and fault tolerant applications.
- ✓ We interact with AWS using Console, CLI, SDK or APIs and automate using IaC tools.
- ✓ Core services include Compute, Storage, Database, Networking, Security, Monitoring and more.
- ✓ These services together help us build scalable, secure and cost-effective solutions.

Learn the building blocks well,
you can build anything on AWS!



INTERVIEW QUESTIONS (BASICS)

- 1 What is AWS?
- 2 What is the difference between Region and Availability Zone?
- 3 Why do we need multiple AZs?
- 4 What is Edge Location and how is it used?
- 5 What is the difference between Authentication and Authorization?
- 6 What is IAM and why is it important?
- 7 How can we interact with AWS?
- 8 Name some core AWS services and their use.
- 9 What is the benefit of using CloudFront?
- 10 What is the main responsibility of AWS in the Shared Responsibility Model?



QUICK REVISION

- Cloud Computing → On-demand resources over internet
- Global Infrastructure → Region → AZ → Edge Location
- Access & Identity → IAM (Users, Groups, Roles, Policies)
- Interact with AWS → Console, CLI, SDK, APIs, IaC
- Core Services → Compute, Storage, Database, Networking, Security, Monitoring & more

Understand the basics deeply,
it makes advanced AWS easy!



AWS LEARNING ROADMAP

ROADMAP
1 PAGE

— STEP BY STEP PATH TO BECOME AWS READY —



This roadmap is for beginners to advanced level.

Follow this path, be consistent and keep building! ☆



1 FOUNDATION (START HERE)

- Understand Cloud Computing basics
- Learn AWS Fundamentals (Regions, AZs, IAM basics, Pricing, Shared Responsibility, Core Services)
- Get comfortable with AWS Management Console

Goal:
Understand the basics and core concepts.



2 IDENTITY & ACCESS

- Deep dive into IAM (Users, Groups, Roles, Policies)
- Learn Security Best Practices
- Understand MFA, Permissions & Least Privilege

Goal:
Secure access to AWS resources.



3 NETWORKING BASICS

- Learn VPC, Subnets, Route Tables, Internet Gateway
- Security Groups vs NACL
- VPC Peering, NAT Gateway, Bastion Host
- (Detailed networking in separate series)

Goal:
Build a secure and isolated network.



4 COMPUTE SERVICES

- Learn EC2 in detail (Instances, AMI, Key Pair, SG)
- Auto Scaling, Elastic Load Balancer
- Understand EC2 pricing & instance types
- Hands-on with user data and launch templates

Goal:
Launch and manage compute resources.



5 STORAGE SERVICES

- Learn S3 (Buckets, Objects, Permissions, Versioning)
- Lifecycle Policies, Storage Classes
- EBS, EFS basics

Goal:
Store and manage data reliably.



6 DATABASE SERVICES

- Learn RDS (MySQL/PostgreSQL)
- Backups, Multi-AZ, Read Replicas
- Introduction to DynamoDB

Goal:
Manage databases in AWS.



7 SECURITY & MONITORING

- AWS Security Best Practices
- CloudWatch (Metrics, Logs, Alarms)
- AWS CloudTrail, Config basics
- KMS, Secrets Manager overview

Goal:
Monitor and secure your environment.



8 ADVANCED & DEVOPS

- Lambda & Serverless basics
- IAM Roles for Services
- Infrastructure as Code (CloudFormation / Terraform)
- CI/CD with AWS (CodePipeline, CodeBuild)
- Containers (ECS / EKS basics)

Goal:
Automate, deploy and scale modern applications.

Practice + Projects + Consistency = AWS Success



☆ LEARNING RESOURCES



AWS Official Training
(aws.amazon.com/training)



AWS Documentation
(docs.aws.amazon.com)



AWS YouTube Channel
(aws.amazon.com/youtube)



Hands-on Labs
(labs.aws, tutorialdojo.com, acloudguru.com)



AWS Community & Forums
(re:Post, Stack Overflow, Dev.to)

TIPS FOR SUCCESS

- ✓ Focus on one topic at a time.
- ✓ Always do hands-on in AWS Free Tier.
- ✓ Build small projects to strengthen your learning.
- ✓ Take notes and keep revising.
- ✓ Understand concepts, not just memorize.
- ✓ Stay consistent and keep learning!

SKILLS YOU WILL BUILD

- Cloud Fundamentals
- Security & Identity Management
- Networking
- Compute & Storage
- Databases
- Monitoring & Logging
- Automation & DevOps
- Cost Optimization



PRACTICE PROJECT IDEAS

- ✓ Host a Static Website using S3 + CloudFront
- ✓ Launch a 3-Tier Web Application (EC2 + RDS + ALB)
- ✓ Build a VPC with Public & Private Subnets
- ✓ Store and Retrieve Files from S3 using IAM Roles
- ✓ Set up Monitoring & Alarms for EC2 using CloudWatch
- ✓ Automate Infrastructure using Terraform
- ✓ Build CI/CD Pipeline for a Web App



KEY TAKEAWAYS ☆

- ★ Start with basics and build strong foundation.
- ★ Hands-on is the key to mastering AWS.
- ★ Understand security from the beginning.
- ★ Learn, Build, Share and Grow with the AWS Community.
- ★ Cloud is the future - Keep learning, Keep building!



"Continuous learning turns beginners into experts."
Keep building on AWS! 😊